Mathematical Methods MTH1021 Problem sheet 20

Line integrals and double integrals. Polar coordinates.

[**SUBMIT ANSWERS TO ALL QUESTIONS**]

Line integral of the first kind: $\int\limits_C f(x,y,z)ds$; second kind: $\int\limits_C \boldsymbol{A}\cdot d\boldsymbol{r} = \int\limits_C (A_x dx + A_y dy + A_z dz)$.

Double integral: $\iint\limits_S f(x,y)dxdy$. Green's theorem: $\oint\limits_C (Pdx + Qdy) = \iint\limits_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy$.

Elements of length and area in polar coordinates (r,θ): $ds = \sqrt{dr^2 + r^2 d\theta^2}$, $dS = r dr d\theta$.

---

1. Find the line integral of the first kind $\int\limits_C xyds$, where C is a quarter of the circle $x^2+y^2=a^2$ lying in the first quadrant. [Hint: use the parameterisation $x = a\cos t$, $y = a\sin t$.] [20]

2. Find $\int\limits_C \frac{ds}{x^2+y^2+z^2}$, where C is the turn of the helix $x = a\cos t$, $y = a\sin t$, $z = bt$ from $t = 0$ to $t = 2\pi$. [Hint: use substitution $u = bt/a$ to obtain a standard integral.] [20]

3. Evaluate the following double integrals:

$$\text{(a)} \int\limits_0^2 dy \int\limits_0^1 (x^2+2y)dx, \quad \text{(b)} \int\limits_3^4 dx \int\limits_1^2 \frac{dy}{(x+y)^2}, \quad \text{(c)} \int\limits_0^{2\pi} d\theta \int\limits_{a\sin\theta}^{a} rdr.$$

[20]

4. Write the equations of curves bounding the integration regions of the following double integrals, and sketch them.

$$\text{(a)} \int\limits_0^4 dx \int\limits_{x/4}^1 f(x,y)dy, \quad \text{(b)} \int\limits_0^3 dx \int\limits_{x^2}^{3x} f(x,y)dy, \quad \text{(c)} \int\limits_0^2 dx \int\limits_0^{\sqrt{9-x^2}} f(x,y)dy,$$

[20]

5. Show that for $P = -y$, $Q = 0$ the integral on the left-hand side of Green's theorem gives the area bounded by curve C. Hence, use this line integral to find the area of the ellipse

$$x = a\cos t, \quad y = b\sin t \quad (0 \le t \le 2\pi).$$

[20]